

Final Exam

PhD Macro

Instructions. Show steps. Partial credit for correct intermediate steps.

Question 1: human capital investment using goods [35 pts]

In the “human capital investment using goods” model (Topic 4, Case 4), the planner chooses next-period stocks (K', X') and consumption C subject to

$$C = F(K, X) + (1 - \delta)K + (1 - \delta_x)X - K' - X',$$

with $\delta \in (0, 1)$ and $\delta_x \in (0, 1)$. Assume Cobb–Douglas technology

$$F(K, X) = K^\alpha X^{1-\alpha}, \quad \alpha \in (0, 1),$$

and CRRA utility

$$u(C) = \frac{C^{1-\sigma} - 1}{1 - \sigma}, \quad \sigma > 0.$$

Let $k \equiv K/X$ and write $F(K, X) = Xf(k)$ where $f(k) = k^\alpha$.

(a) **BGP condition for k^* .**

[14 pts]

Using the slide result that an interior BGP with $K' > 0$ and $X' > 0$ implies

$$f'(k_{t+1}) - \delta = f(k_{t+1}) - k_{t+1}f'(k_{t+1}) - \delta_x,$$

and imposing $k_{t+1} = k_t = k^*$ on the BGP, show that k^* solves

$$(1 + k)f'(k) - f(k) = \delta - \delta_x. \quad (3)$$

For Cobb–Douglas $f(k) = k^\alpha$, show that when $\delta = \delta_x$ the unique solution is

$$k^* = \frac{\alpha}{1 - \alpha}.$$

(If $\delta \neq \delta_x$, k^* is obtained by solving (3) numerically, as done in the slides.)

(b) **Growth rate of growing series.**

[10 pts]

Using the slide formula for the growth rate of growing series,

$$\gamma_Y = \cdots = \gamma_C = [\beta (f'(k^*) + 1 - \delta)]^{1/\sigma},$$

compute γ_Y for

$$\alpha = 0.30, \quad \delta = 0.10, \quad \delta_x = 0.15, \quad \beta = 0.985, \quad \sigma = 2,$$

using the slide-reported solution $k^* = 0.42857$.

- (c) **Interpretation (one line).** [11 pts]
Holding $(\alpha, \beta, \sigma, \delta)$ fixed, what is the direction of change in k^* when δ_x increases?
Briefly explain using equation (3).

Solution. (a) Rearranging the slide equation gives:

$$f'(k_{t+1}) + k_{t+1}f'(k_{t+1}) - f(k_{t+1}) = \delta - \delta_x \Rightarrow (1 + k_{t+1})f'(k_{t+1}) - f(k_{t+1}) = \delta - \delta_x.$$

On the BGP, $k_{t+1} = k_t = k^*$, yielding (3). For Cobb–Douglas, $f'(k) = \alpha k^{\alpha-1}$. If $\delta = \delta_x$, (3) becomes

$$(1 + k)\alpha k^{\alpha-1} - k^\alpha = 0 \Rightarrow k^{\alpha-1}(\alpha - (1 - \alpha)k) = 0 \Rightarrow k^* = \frac{\alpha}{1 - \alpha}.$$

(b) With $k^* = 0.42857$, $f'(k^*) = \alpha(k^*)^{\alpha-1} = 0.30 \cdot (0.42857)^{-0.70} \approx 0.543$. Then $f'(k^*) + 1 - \delta \approx 0.543 + 0.90 = 1.443$. So

$$\gamma_Y \approx [0.985 \cdot 1.443]^{1/2} \approx [1.421]^{1/2} \approx 1.192.$$

(c) Equation (3) has RHS $\delta - \delta_x$. Increasing δ_x lowers RHS. In the Cobb–Douglas case, the solution shifts so that K/X rises: faster depreciation of X makes X relatively harder to accumulate, so the balanced ratio features higher $k^* = K/X$.

Question 2: Lucas (1988) human capital accumulation model [30 pts]

In the Lucas (1988) model (Topic 4), specialize production to

$$Y_t = K_t^\alpha M_t^{1-\alpha}, \quad \alpha \in (0, 1),$$

where $M_t = \phi_t X_t$, X_t is the stock of skills, and $\phi_t \in (0, 1)$ is the fraction of unit time endowment spent working. The fraction $1 - \phi_t$ is time in education. On a balanced growth path with constant ϕ , skills grow according to the slide equation:

$$\gamma_X \equiv \frac{X_{t+1}}{X_t} = 1 + A(1 - \phi), \quad A > 0.$$

Preferences are CRRA with parameter σ and discount factor β .

- (a) **BGP growth rate.** [12 pts]
Using the slide result, state and use the key implication:

$$\gamma_Y = \cdots = \gamma_X = [\beta(1 + A)]^{1/\sigma}.$$

- (b) **Solve for ϕ .** [10 pts]
Combine $\gamma_X = 1 + A(1 - \phi)$ with the BGP growth rate to show:

$$\phi = \frac{1}{A} \left[1 + A - (\beta(1 + A))^{1/\sigma} \right].$$

(c) **Numerical.**

[8 pts]

Compute ϕ , γ_X , and γ_Y when

$$\beta = 0.96, \quad \sigma = 2, \quad A = 0.20.$$

Solution. (a) Topic 4 derives: $\gamma_Y = \dots = \gamma_X = [\beta(1 + A)]^{1/\sigma}$.

(b) Solve $1 + A(1 - \phi) = [\beta(1 + A)]^{1/\sigma}$:

$$A(1 - \phi) = [\beta(1 + A)]^{1/\sigma} - 1 \Rightarrow \phi = \frac{1}{A} [1 + A - [\beta(1 + A)]^{1/\sigma}].$$

(c) $[\beta(1 + A)]^{1/\sigma} = [0.96 \cdot 1.20]^{1/2} = [1.152]^{1/2} \approx 1.073$. Then $\phi = \frac{1}{0.20}(1.20 - 1.073) = 5(0.127) \approx 0.635$. So $\gamma_X = 1 + A(1 - \phi) = 1 + 0.20(0.365) \approx 1.073$, and $\gamma_Y = \gamma_X \approx 1.073$.

Question 3: aggregation over heterogenous firms and $A(\mu)$

[35 pts]

There is a large number of heterogeneous firms. Production at each firm is

$$y = z\varepsilon k^\alpha n^\nu, \tag{1}$$

where $\alpha \in (0, 1)$, $\nu \in (0, 1)$, and $\alpha + \nu < 1$. Capital is rented and labor is hired. Profits are

$$\pi(z\varepsilon; r, w) \equiv \max_{k, n} z\varepsilon k^\alpha n^\nu - (r + \delta)k - wn, \tag{2}$$

where r is the real interest rate, w is the real wage, and $\delta \in (0, 1)$. Optimal choices satisfy

$$\alpha z\varepsilon k^{\alpha-1} n^\nu = r + \delta, \quad \nu z\varepsilon k^\alpha n^{\nu-1} = w.$$

Let $\mu(\varepsilon)$ describe the measure of firms in production over ε .

(a) **Firm choices (equations (6)–(8)).**

[12 pts]

Dividing (3) by (4) gives equation (6):

$$\frac{\alpha}{\nu} \frac{n}{k} = \frac{r + \delta}{w}. \tag{6}$$

Using (6) in (3), show that capital demand is (equation (7) in the slides):

$$k(\varepsilon) = \left(\frac{z\varepsilon \alpha^{1-\nu} \nu^\nu}{(r + \delta)^{1-\nu} w^\nu} \right)^{\frac{1}{1-(\alpha+\nu)}}. \tag{7}$$

(You do not need to re-derive the labor demand formula (8).)

(b) **Market clearing and the key moment.**

[12 pts]

Using the slide market-clearing equations (10)–(11),

$$K = \kappa_K(z, r + \delta, w) \int \varepsilon^{\frac{1}{1-(\alpha+\nu)}} \mu(d\varepsilon), \quad (10)$$

$$N = \kappa_N(z, r + \delta, w) \int \varepsilon^{\frac{1}{1-(\alpha+\nu)}} \mu(d\varepsilon), \quad (11)$$

define

$$\mathcal{M}(\mu) \equiv \int \varepsilon^{\frac{1}{1-(\alpha+\nu)}} \mu(d\varepsilon).$$

Explain why, holding $(z, r + \delta, w)$ fixed, changes in μ affect K and N only through $\mathcal{M}(\mu)$.

(c) **Distribution changes and aggregate productivity.**

[11 pts]

Define an “endogenous TFP” index

$$A(\mu) \equiv (\mathcal{M}(\mu))^{1-(\alpha+\nu)}.$$

Let $\alpha = 0.30$, $\nu = 0.50$ so $1 - (\alpha + \nu) = 0.20$. Compare:

$$\mu_1 : \varepsilon = 1 \text{ w.p. } 1, \quad \mu_2 : \varepsilon = 0.5 \text{ w.p. } 0.5, \varepsilon = 1.5 \text{ w.p. } 0.5.$$

Compute $A(\mu_1)$ and $A(\mu_2)$. Does greater dispersion raise or lower $A(\mu)$ here?

Solution. (a) This is exactly equation (7) in production1b.

(b) In (10)–(11), both K and N depend on μ only through the same integral moment $\mathcal{M}(\mu)$. Hence shifts in the distribution that change $\mathcal{M}(\mu)$ shift aggregate inputs (for fixed prices) and, equivalently, shift measured productivity in the aggregate production function.

(c) Here $\frac{1}{1-(\alpha+\nu)} = \frac{1}{0.20} = 5$, so $\mathcal{M}(\mu) = \mathbb{E}[\varepsilon^5]$ and $A(\mu) = \mathcal{M}(\mu)^{0.20}$. For μ_1 : $\mathcal{M} = 1 \Rightarrow A(\mu_1) = 1$. For μ_2 : $\mathcal{M} = 0.5(0.5^5) + 0.5(1.5^5) = 0.5(0.03125 + 7.59375) = 3.8125$. So $A(\mu_2) = 3.8125^{0.20} \approx e^{0.20 \ln 3.8125} \approx e^{0.268} \approx 1.307 > 1$. Thus more dispersion raises $A(\mu)$ in this example (convexity of ε^5).